KERALA POLYTECHNIC · REVISION 2026

Digital Computer Fundamentals

Complete student lesson for course code **3134**.

Revision 2026 Semester 3 Program Core 4 modules

Course snapshot

Scheme: 3 (L:3,T:0,P:0); 45 periods

Credits: 3

Contact: 3 (L:3,T:0,P:0)

Module hours listed: 43

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Handbook method: Do not treat a topic as completed after reading its heading. For every topic card, complete the learning objectives, follow the conceptual framework, write the worked answer method, and answer the self-check questions. This creates a study record that is useful for class learning and examination preparation.

Course Dashboard

Course code

3134

Semester

3

Credits

3

Teaching scheme

3 (L:3,T:0,P:0); 45 periods

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Number systems and digital codes	8	CO1 · Applying
2	Boolean algebra and Karnaugh maps	11	CO2 · Applying

No.	Module	Official hours	Relevant CO / level
3	Combinational circuits	12	CO3 · Applying
4	Sequential circuits	12	CO4 · Applying

Prerequisites

- Fundamentals of Electrical and Electronics; semiconductor physics, diodes, transistors and rectifiers.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. Understand the data representation in the Computer System.
2. Perform number conversion from one system to another.
3. Understand use of boolean algebra and K-Map in digital circuits.
4. Equip students to design and develop simple combinational and sequential circuits.

Course Outcomes

1. Perform number system conversions, binary arithmetic operations and binary coding.
2. Make use of Boolean algebra and the Karnaugh Map for the implementation of logic functions.
3. Design combinational circuits.
4. Design sequential circuits.

CO-PO mapping as supplied

CO–PO mapping is reproduced from the supplied syllabus header; 3 = strongly mapped, 2 = moderately mapped, 1 = weakly mapped, and a blank cell is not silently converted into a value.

Complete Syllabus Coverage

Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

Source-to-lesson coverage map

Module	Lesson topic cards	Source basis	Official point units
Module 1	3	Official topic-row context	5
Module 2	3	Official topic-row context	5
Module 3	4	Official topic-row context	4
Module 4	4	Official topic-row context	4

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Number systems and base conversion	3
module-1	M1.02	Complements and binary arithmetic	3
module-1	M1.03	Binary codes and data representation	2
module-2	M2.01	Boolean laws and logic gates	4
module-2	M2.02	Canonical forms, minterms and maxterms	3
module-2	M2.03	Karnaugh-map simplification	4
module-3	M3.01	Half and full adders	3
module-3	M3.02	Subtractors and arithmetic circuits	3

Module	Topic code	Topic	Hours
module-3	M3.03	Encoders, decoders and comparators	3
module-3	M3.04	Multiplexers and demultiplexers	3
module-4	M4.01	Latches and flip-flops	3
module-4	M4.02	Registers and shift registers	3
module-4	M4.03	Counters and frequency division	3
module-4	M4.04	Timing and sequential-system design	3

Learning Roadmap

1. Number systems and digital codes
CO1 · Applying · 8 hours

2. Boolean algebra and Karnaugh maps
CO2 · Applying · 11 hours

3. Combinational circuits
CO3 · Applying · 12 hours

4. Sequential circuits
CO4 · Applying · 12 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Number systems and digital codes

Data representation is the foundation for digital hardware. The official content covers decimal, binary, octal and hexadecimal systems; conversions; signed and unsigned representations; complements; binary arithmetic; BCD, Gray and ASCII codes.

Hours: 8
CO1 · Applying · Bloom level: Applying

Handbook study routine: Complete Module 1 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 1

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying
M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding
M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding
M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying
M1.05 Solve addition of BCD numbers 1 Applying Contents:

Point-by-point study guide

Official syllabus point 1: M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying

Exact source point

Syllabus text: M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying using the official terminology retained above.
- Organise the important elements of M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying into a logical sequence, classification, structure, or calculation.
- Connect M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying?
- How would you distinguish M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying from a closely related idea?
- Where would an engineer or technician encounter M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.

- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M1.01 2 Applying system todifferent number systems Apply arithmetic operations on binary M1.02 2 Applying □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 2: M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding

Exact source point

Syllabus text: M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding

How to understand and study it

This point is an official syllabus requirement: “M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□□ M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding using the official terminology retained above.
- Organise the important elements of M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.
Required: the logic sequence, program structure, truth table, or operating result.
Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding?
- How would you distinguish M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding from a closely related idea?
- Where would an engineer or technician encounter M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M1.02 2 Applying numbers M1.03 Represent signed numbers in digital systems 1 Understanding □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 3: M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding

Exact source point

Syllabus text: M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding

How to understand and study it

This point is an official syllabus requirement: “M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding □□□□ technical terms English-□□□□□□□ □□□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding using the official terminology retained above.
- Organise the important elements of M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.
Required: the logic sequence, program structure, truth table, or operating result.
Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding?
- How would you distinguish M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding from a closely related idea?
- Where would an engineer or technician encounter M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M1.03 Represent signed numbers in digital systems 1 Understanding M1.04 Summarize different types of binary codes 2 Understanding □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 4: M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying

Exact source point

> **Syllabus text:** M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying

How to understand and study it

This point is an official syllabus requirement: “M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying using the official terminology retained above.
- Organise the important elements of M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying into a logical sequence, classification, structure, or calculation.
- Connect M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying?
- How would you distinguish M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying from a closely related idea?
- Where would an engineer or technician encounter M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.

- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.04 Summarize different types of binary codes 2 Understanding M1.05 Solve addition of BCD numbers 1 Applying □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 5: M1.05 Solve addition of BCD numbers 1 Applying Contents

Exact source point

Syllabus text: M1.05 Solve addition of BCD numbers 1 Applying Contents

How to understand and study it

This point is an official syllabus requirement: “M1.05 Solve addition of BCD numbers 1 Applying Contents”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ M1.05 Solve addition of BCD numbers 1 Applying Contents □□□□ technical terms English-□□□□□□□□ □□□□□□□□. □□□□□□ definition □□□□□□□□□□□ principle □□□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Diagram, sequence, formula, unit □□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□ revision □□□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.05 Solve addition of BCD numbers 1 Applying Contents is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.05 Solve addition of BCD numbers 1 Applying Contents using the official terminology retained above.
- Organise the important elements of M1.05 Solve addition of BCD numbers 1 Applying Contents into a logical sequence, classification, structure, or calculation.
- Connect M1.05 Solve addition of BCD numbers 1 Applying Contents with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.05 Solve addition of BCD numbers 1 Applying Contents with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.05 Solve addition of BCD numbers 1 Applying Contents. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.05 Solve addition of BCD numbers 1 Applying Contents is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.05 Solve addition of BCD numbers 1 Applying Contents, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.05 Solve addition of BCD numbers 1 Applying Contents, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.05 Solve addition of BCD numbers 1 Applying Contents?
- How would you distinguish M1.05 Solve addition of BCD numbers 1 Applying Contents from a closely related idea?
- Where would an engineer or technician encounter M1.05 Solve addition of BCD numbers 1 Applying Contents, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.05 Solve addition of BCD numbers 1 Applying Contents incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

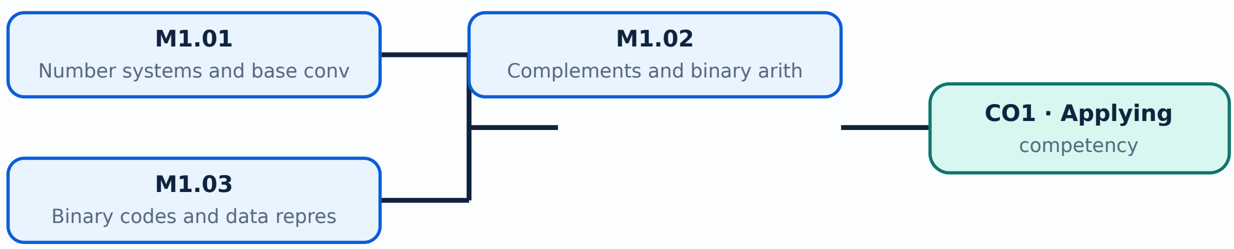
Malayalam support: M1.05 Solve addition of BCD numbers 1 Applying Contents □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Learning goals

- Convert numbers between common bases.
- Perform binary arithmetic and identify suitable digital codes.
- Check a conversion by reversing the operation.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Number systems and base conversion **(3 hours)**

Concept and explanation

A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Use positional weights.
- Convert integer and fractional values carefully.
- Write the radix when ambiguity is possible.

> **Engineering / workplace application:** Microprocessors and memory addresses commonly use hexadecimal notation while logic circuits use binary states.

Handbook treatment

M1.01 — Number systems and base conversion (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.01 — Number systems and base conversion (3 hours) using the official terminology retained above.
- Organise the important elements of M1.01 — Number systems and base conversion (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.01 — Number systems and base conversion (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.01 — Number systems and base conversion (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.01 — Number systems and base conversion (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.01 — Number systems and base conversion (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.01 — Number systems and base conversion (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.01 — Number systems and base conversion (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 — Number systems and base conversion (3 hours)?
- How would you distinguish M1.01 — Number systems and base conversion (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.01 — Number systems and base conversion (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 — Number systems and base conversion (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.01 — Number systems and base conversion (3 hours)
[illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M1.02 — Complements and binary arithmetic (3 hours)

Concept and explanation

One’s complement and two’s complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Distinguish one’s and two’s complement.
- Check overflow in fixed-width arithmetic.
- Keep the word length consistent.

Illustrative example: For an n-bit two’s-complement number, negate it by complementing all bits and adding 1, then discard a carry beyond the word length.

Handbook treatment

M1.02 — Complements and binary arithmetic (3 hours) must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M1.02 — Complements and binary arithmetic (3 hours) using the official terminology retained above.
- Organise the important elements of M1.02 — Complements and binary arithmetic (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.02 — Complements and binary arithmetic (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.02 — Complements and binary arithmetic (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.02 — Complements and binary arithmetic (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M1.02 — Complements and binary arithmetic (3 hours) is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M1.02 — Complements and binary arithmetic (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.02 — Complements and binary arithmetic (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 — Complements and binary arithmetic (3 hours)?
- How would you distinguish M1.02 — Complements and binary arithmetic (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.02 — Complements and binary arithmetic (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 — Complements and binary arithmetic (3 hours) incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M1.02 — Complements and binary arithmetic (3 hours)
□□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

M1.03 — Binary codes and data representation (2 hours)

Concept and explanation

BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Explain BCD, Gray and ASCII.
- Convert a small value into a code.
- State one engineering use for each code.

> **Engineering / workplace application:** Encoders, displays, keyboards and digital communication interfaces use coded representations.

Handbook treatment

M1.03 — Binary codes and data representation (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.03 — Binary codes and data representation (2 hours) using the official terminology retained above.
- Organise the important elements of M1.03 — Binary codes and data representation (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.03 — Binary codes and data representation (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Number systems and digital codes.
- Write an examination answer on M1.03 — Binary codes and data representation (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.03 — Binary codes and data representation (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.03 — Binary codes and data representation (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.03 — Binary codes and data representation (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.03 — Binary codes and data representation (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 — Binary codes and data representation (2 hours)?
- How would you distinguish M1.03 — Binary codes and data representation (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.03 — Binary codes and data representation (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 — Binary codes and data representation (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.03 — Binary codes and data representation (2 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: A reliable conversion method prevents errors when a digital quantity is represented in another base.

OFFICIAL MODULE 2

Boolean algebra and Karnaugh maps

Boolean algebra models logic with variables that take 0 or 1. The syllabus includes Boolean laws, logic gates, canonical forms, minterms, maxterms and Karnaugh-map simplification.

Hours: 11
CO2 · Applying · Bloom level: Applying

Handbook study routine: Complete Module 2 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 2

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3
M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding
M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1
M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying
M2.05 2 Applying logic functions Series Test - I 1

Point-by-point study guide

Official syllabus point 1: M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3

Exact source point

Syllabus text: M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3

How to understand and study it

This point is an official syllabus requirement: “M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 using the official terminology retained above.
- Organise the important elements of M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 into a logical sequence, classification, structure, or calculation.
- Connect M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3?
- How would you distinguish M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 from a closely related idea?
- Where would an engineer or technician encounter M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M2.01 Illustrate basic theorems of Boolean algebra 2 Understanding Make use of Boolean algebra to simplify Applying M2.02 3 [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 2: M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding

Exact source point

> **Syllabus text:** M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding

How to understand and study it

This point is an official syllabus requirement: “M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ M2.02 3 logic functions M2.03 Illustrate properties of basic, universal gates 3 Understanding □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding using the official terminology retained above.
- Organise the important elements of M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.
Required: the logic sequence, program structure, truth table, or operating result.
Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding?
- How would you distinguish M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding from a closely related idea?
- Where would an engineer or technician encounter M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M2.02 3 logic functions M2.03 Illustrate properties of basic and universal gates 3 Understanding □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 3: M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1

Exact source point

Syllabus text: M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1

How to understand and study it

This point is an official syllabus requirement: “M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ M2.03 Illustrate properties of basic, universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 using the official terminology retained above.
- Organise the important elements of M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 into a logical sequence, classification, structure, or calculation.
- Connect M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.
Required: the logic sequence, program structure, truth table, or operating result.
Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1?
- How would you distinguish M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 from a closely related idea?
- Where would an engineer or technician encounter M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M2.03 Illustrate properties of basic and universal gates 3 Understanding Make use of logic gates to implement logic Applying M2.04 1 □□□□ topic □□□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 4: M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying

Exact source point

Syllabus text: M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying using the official terminology retained above.
- Organise the important elements of M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying into a logical sequence, classification, structure, or calculation.
- Connect M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying?
- How would you distinguish M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying from a closely related idea?
- Where would an engineer or technician encounter M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.04 1 functions Make use of K-Map for the implementation of M2.05 2 Applying □□□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□□ definition □□□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□□.

Official syllabus point 5: M2.05 2 Applying logic functions Series Test - I 1

Exact source point

Syllabus text: M2.05 2 Applying logic functions Series Test - I 1

How to understand and study it

This point is an official syllabus requirement: “M2.05 2 Applying logic functions Series Test - I 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ M2.05 2 Applying logic functions Series Test - I 1 □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.05 2 Applying logic functions Series Test - I 1 should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M2.05 2 Applying logic functions Series Test - I 1 using the official terminology retained above.
- Organise the important elements of M2.05 2 Applying logic functions Series Test - I 1 into a logical sequence, classification, structure, or calculation.
- Connect M2.05 2 Applying logic functions Series Test - I 1 with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.05 2 Applying logic functions Series Test - I 1 with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.05 2 Applying logic functions Series Test - I 1. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M2.05 2 Applying logic functions Series Test - I 1 matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.
Required: the logic sequence, program structure, truth table, or operating result.
Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M2.05 2 Applying logic functions Series Test - I 1, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.05 2 Applying logic functions Series Test - I 1, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.05 2 Applying logic functions Series Test - I 1?
- How would you distinguish M2.05 2 Applying logic functions Series Test - I 1 from a closely related idea?
- Where would an engineer or technician encounter M2.05 2 Applying logic functions Series Test - I 1, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.05 2 Applying logic functions Series Test - I 1 incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

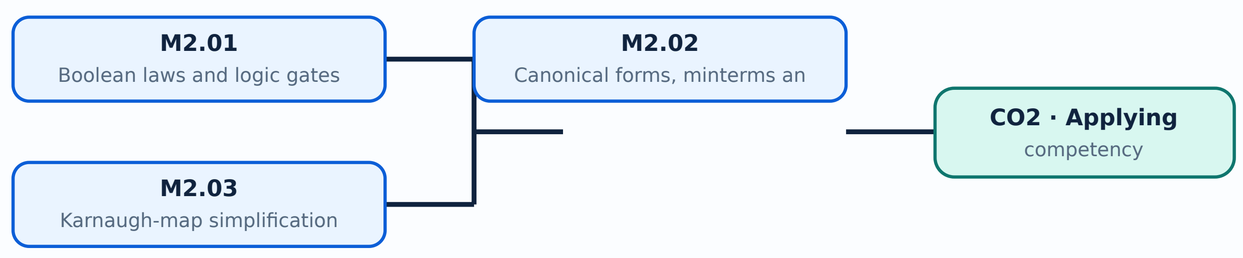
Malayalam support: M2.05 2 Applying logic functions Series Test - I 1 □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□□.

Learning goals

- Apply Boolean identities.
- Reduce a logic expression using a K-map.
- Translate between truth tables and expressions.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Boolean laws and logic gates (4 hours)

Concept and explanation

Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan’s theorems are especially useful for changing a circuit between active-high and active-low forms.

Malayalam support: □ □□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Use identity, null, idempotent and complement laws.
- Relate NAND and NOR to universal-gate implementation.
- Draw a truth table.

Engineering / workplace application: Combinational logic in controllers and interlocks is specified using Boolean expressions.

Handbook treatment

M2.01 — Boolean laws and logic gates (4 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M2.01 — Boolean laws and logic gates (4 hours) using the official terminology retained above.
- Organise the important elements of M2.01 — Boolean laws and logic gates (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.01 — Boolean laws and logic gates (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.01 — Boolean laws and logic gates (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.01 — Boolean laws and logic gates (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M2.01 — Boolean laws and logic gates (4 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.
Required: the logic sequence, program structure, truth table, or operating result.
Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M2.01 — Boolean laws and logic gates (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.01 — Boolean laws and logic gates (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 — Boolean laws and logic gates (4 hours)?
- How would you distinguish M2.01 — Boolean laws and logic gates (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.01 — Boolean laws and logic gates (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 — Boolean laws and logic gates (4 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M2.01 — Boolean laws and logic gates (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M2.02 — Canonical forms, minterms and maxterms (3 hours)

Concept and explanation

A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□□ □□□□□□□□□□.

Study points

- Identify minterms from rows containing 1.
- Identify maxterms from rows containing 0.
- Use don’t-care conditions only when justified.

> **Common mistake:** Do not interchange the row conditions for minterms and maxterms.

Handbook treatment

M2.02 — Canonical forms, minterms and maxterms (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.02 — Canonical forms, minterms and maxterms (3 hours) using the official terminology retained above.
- Organise the important elements of M2.02 — Canonical forms, minterms and maxterms (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.02 — Canonical forms, minterms and maxterms (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.02 — Canonical forms, minterms and maxterms (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.02 — Canonical forms, minterms and maxterms (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.02 — Canonical forms, minterms and maxterms (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.02 — Canonical forms, minterms and maxterms (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.02 — Canonical forms, minterms and maxterms (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 — Canonical forms, minterms and maxterms (3 hours)?
- How would you distinguish M2.02 — Canonical forms, minterms and maxterms (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.02 — Canonical forms, minterms and maxterms (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 — Canonical forms, minterms and maxterms (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.02 — Canonical forms, minterms and maxterms (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M2.03 — Karnaugh-map simplification (4 hours)

Concept and explanation

A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.

Malayalam support: □ □□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Make valid groups.
- Select essential prime implicants.
- Implement the reduced expression with suitable gates.

Illustrative example: For four variables, begin with the largest valid group, then cover every required 1-cell with the fewest essential groups.

Handbook treatment

M2.03 — Karnaugh-map simplification (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.03 — Karnaugh-map simplification (4 hours) using the official terminology retained above.
- Organise the important elements of M2.03 — Karnaugh-map simplification (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.03 — Karnaugh-map simplification (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Boolean algebra and Karnaugh maps.
- Write an examination answer on M2.03 — Karnaugh-map simplification (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.03 — Karnaugh-map simplification (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.03 — Karnaugh-map simplification (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.03 — Karnaugh-map simplification (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.03 — Karnaugh-map simplification (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 — Karnaugh-map simplification (4 hours)?
- How would you distinguish M2.03 — Karnaugh-map simplification (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.03 — Karnaugh-map simplification (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 — Karnaugh-map simplification (4 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.03 — Karnaugh-map simplification (4 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Module summary: Simplification reduces gate count, delay and wiring in a practical digital circuit.

OFFICIAL MODULE 3

Combinational circuits

Combinational outputs depend only on present inputs. The official coverage includes adders, subtractors, comparators, encoders, decoders, multiplexers and demultiplexers.

Hours: 12
CO3 · Applying · Bloom level: Applying

Handbook study routine: Complete Module 3 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 3

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying
M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying
M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for
M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding

Point-by-point study guide

Official syllabus point 1: M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying

Exact source point

Syllabus text: M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ M3.01 2 Understanding circuits M3.02 Construct half adder, full adder circuits 2 Applying □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying using the official terminology retained above.
- Organise the important elements of M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying into a logical sequence, classification, structure, or calculation.
- Connect M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying?
- How would you distinguish M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying from a closely related idea?
- Where would an engineer or technician encounter M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying, and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.01 2 Understanding circuits M3.02 Construct half adder and full adder circuits 2 Applying □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 2: M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying

Exact source point

Syllabus text: M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying

How to understand and study it

This point is an official syllabus requirement: “M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ M3.02 Construct half adder, full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying using the official terminology retained above.
- Organise the important elements of M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying into a logical sequence, classification, structure, or calculation.
- Connect M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying?
- How would you distinguish M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying from a closely related idea?
- Where would an engineer or technician encounter M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying, and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.02 Construct half adder and full adder circuits 2 Applying Make use of binary parallel adder toimplement M3.03 2 Applying □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Official syllabus point 3: M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for

Exact source point

> **Syllabus text:** M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for

How to understand and study it

This point is an official syllabus requirement: “M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for using the official terminology retained above.
- Organise the important elements of M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for into a logical sequence, classification, structure, or calculation.
- Connect M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for?
- How would you distinguish M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for from a closely related idea?
- Where would an engineer or technician encounter M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for, and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 2 Applying parallel adder-subtractor circuit Construct combinational circuits for incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.03 2 Applying parallel adder-subtractor circuit
Construct combinational circuits for [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 4: M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding

Exact source point

> **Syllabus text:** M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding

How to understand and study it

This point is an official syllabus requirement: “M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding”. Prepare a comparison using the same criteria for every item: principle, performance, cost or complexity, limitation, and application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding using the official terminology retained above.
- Organise the important elements of M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding into a logical sequence, classification, structure, or calculation.
- Connect M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.
Required: the logic sequence, program structure, truth table, or operating result.
Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding?
- How would you distinguish M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding from a closely related idea?
- Where would an engineer or technician encounter M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

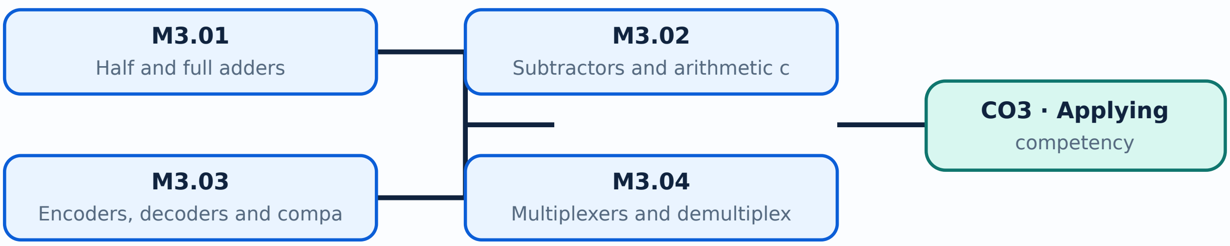
Malayalam support: M3.04 multiplication,comparison, multiplexing, 6 Applying decoding, encoding and priority encoding ▯▯▯▯ topic ▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ exact technical term ▯▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯▯ principle, named parts/steps, application, limitation ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. English terminology, symbols, units, diagram labels ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯. Exam answer ▯▯▯▯▯▯▯▯▯▯▯ concept-▯▯▯▯ ▯▯▯▯▯–▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯.

Learning goals

- Construct a truth table and logic expression.
- Choose a standard combinational block.
- Verify the output for representative input combinations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

− M3.01 — Half and full adders (3 hours)

Concept and explanation

A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.

Malayalam support: □ □□□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Write sum and carry equations.
- Explain carry propagation.
- Distinguish half and full adders.

Engineering / workplace application: Arithmetic logic units use cascaded full adders.

Handbook treatment

M3.01 — Half and full adders (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.01 — Half and full adders (3 hours) using the official terminology retained above.
- Organise the important elements of M3.01 — Half and full adders (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.01 — Half and full adders (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.01 — Half and full adders (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.01 — Half and full adders (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.01 — Half and full adders (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.01 — Half and full adders (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.01 — Half and full adders (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 — Half and full adders (3 hours)?
- How would you distinguish M3.01 — Half and full adders (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.01 — Half and full adders (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 — Half and full adders (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.01 — Half and full adders (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.02 — Subtractors and arithmetic circuits (3 hours)

Concept and explanation

Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- State difference and borrow outputs.
- Explain two’s-complement subtraction.
- Check sign and overflow conditions.

Handbook treatment

M3.02 — Subtractors and arithmetic circuits (3 hours) must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M3.02 — Subtractors and arithmetic circuits (3 hours) using the official terminology retained above.
- Organise the important elements of M3.02 — Subtractors and arithmetic circuits (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.02 — Subtractors and arithmetic circuits (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.02 — Subtractors and arithmetic circuits (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.02 — Subtractors and arithmetic circuits (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M3.02 — Subtractors and arithmetic circuits (3 hours) is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M3.02 — Subtractors and arithmetic circuits (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.02 — Subtractors and arithmetic circuits (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 — Subtractors and arithmetic circuits (3 hours)?
- How would you distinguish M3.02 — Subtractors and arithmetic circuits (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.02 — Subtractors and arithmetic circuits (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.02 — Subtractors and arithmetic circuits (3 hours) incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M3.02 — Subtractors and arithmetic circuits (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.03 — Encoders, decoders and comparators (3 hours)

Concept and explanation

An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.

Malayalam support: □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Identify input/output roles.
- Use enable inputs correctly.
- Interpret priority encoders where more than one input may be active.

Engineering / workplace application: Address decoding and keypad interfaces use decoders and encoders.

Handbook treatment

M3.03 — Encoders, decoders and comparators (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.03 — Encoders, decoders and comparators (3 hours) using the official terminology retained above.
- Organise the important elements of M3.03 — Encoders, decoders and comparators (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.03 — Encoders, decoders and comparators (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.03 — Encoders, decoders and comparators (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.03 — Encoders, decoders and comparators (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.03 — Encoders, decoders and comparators (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.03 — Encoders, decoders and comparators (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.03 — Encoders, decoders and comparators (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 — Encoders, decoders and comparators (3 hours)?
- How would you distinguish M3.03 — Encoders, decoders and comparators (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.03 — Encoders, decoders and comparators (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 — Encoders, decoders and comparators (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.03 — Encoders, decoders and comparators (3 hours)
[illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M3.04 — Multiplexers and demultiplexers (3 hours)

Concept and explanation

A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.

Malayalam support: □ □□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Calculate the number of select lines.
- Trace the selected data path.
- Use a multiplexer to realise a Boolean function.

Handbook treatment

M3.04 — Multiplexers and demultiplexers (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.04 — Multiplexers and demultiplexers (3 hours) using the official terminology retained above.
- Organise the important elements of M3.04 — Multiplexers and demultiplexers (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.04 — Multiplexers and demultiplexers (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Combinational circuits.
- Write an examination answer on M3.04 — Multiplexers and demultiplexers (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.04 — Multiplexers and demultiplexers (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.04 — Multiplexers and demultiplexers (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.04 — Multiplexers and demultiplexers (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.04 — Multiplexers and demultiplexers (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 — Multiplexers and demultiplexers (3 hours)?
- How would you distinguish M3.04 — Multiplexers and demultiplexers (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.04 — Multiplexers and demultiplexers (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 — Multiplexers and demultiplexers (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.04 — Multiplexers and demultiplexers (3 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: A clean design proceeds from specification to truth table, simplification, circuit drawing and verification.

OFFICIAL MODULE 4

Sequential circuits

Sequential circuits include memory, so output depends on present inputs and previous state. The syllabus covers latches, flip-flops, registers and counters.

Hours: 12
CO4 · Applying · Bloom level: Applying

Handbook study routine: Complete Module 4 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 4

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops
M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding
M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying
M4.04 6 Applying counters using flip flops Series Test - II 1

Point-by-point study guide

Official syllabus point 1: M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops

Exact source point

Syllabus text: M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops

How to understand and study it

This point is an official syllabus requirement: “M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□□ M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches, M4.02 4 Applying flip flops □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops using the official terminology retained above.
- Organise the important elements of M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops into a logical sequence, classification, structure, or calculation.
- Connect M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops?
- How would you distinguish M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops from a closely related idea?
- Where would an engineer or technician encounter M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops, and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.01 Explain sequential circuits 1 Understanding Illustrate operation of different latches and M4.02 4 Applying flip flops □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Official syllabus point 2: M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding

Exact source point

Syllabus text: M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding

How to understand and study it

This point is an official syllabus requirement: “M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding using the official terminology retained above.
- Organise the important elements of M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding?
- How would you distinguish M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding from a closely related idea?
- Where would an engineer or technician encounter M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.02 4 Applying flip flops M4.03 Outline different type of registers 1 Understanding [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 3: M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying

Exact source point

Syllabus text: M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying

How to understand and study it

This point is an official syllabus requirement: “M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: □ syllabus point □□□□□□□□□□□ M4.03 Outline different type of registers 1 Understanding Construct asynchronous, synchronous M4.04 6 Applying □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying using the official terminology retained above.
- Organise the important elements of M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying into a logical sequence, classification, structure, or calculation.
- Connect M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying?
- How would you distinguish M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying from a closely related idea?
- Where would an engineer or technician encounter M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying, and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.

- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.03 Outline different type of registers 1 Understanding Construct asynchronous and synchronous M4.04 6 Applying [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

Official syllabus point 4: M4.04 6 Applying counters using flip flops Series Test - II 1

Exact source point

Syllabus text: M4.04 6 Applying counters using flip flops Series Test - II 1

How to understand and study it

This point is an official syllabus requirement: “M4.04 6 Applying counters using flip flops Series Test - II 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ▯ syllabus point ▯▯▯▯▯▯▯▯▯▯▯ M4.04 6 Applying counters using flip flops Series Test - II 1 ▯▯▯▯ technical terms English-▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯ principle ▯▯▯▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯ ▯▯▯ term-▯▯▯▯ function, difference, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Diagram, sequence, formula, unit ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ revision ▯▯▯▯▯▯▯.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.04 6 Applying counters using flip flops Series Test - II 1 is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.04 6 Applying counters using flip flops Series Test - II 1 using the official terminology retained above.
- Organise the important elements of M4.04 6 Applying counters using flip flops Series Test - II 1 into a logical sequence, classification, structure, or calculation.
- Connect M4.04 6 Applying counters using flip flops Series Test - II 1 with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.04 6 Applying counters using flip flops Series Test - II 1 with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.04 6 Applying counters using flip flops Series Test - II 1. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.04 6 Applying counters using flip flops Series Test - II 1 is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.04 6 Applying counters using flip flops Series Test - II 1, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.04 6 Applying counters using flip flops Series Test - II 1, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 6 Applying counters using flip flops Series Test - II 1?
- How would you distinguish M4.04 6 Applying counters using flip flops Series Test - II 1 from a closely related idea?
- Where would an engineer or technician encounter M4.04 6 Applying counters using flip flops Series Test - II 1, and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 6 Applying counters using flip flops Series Test - II 1 incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

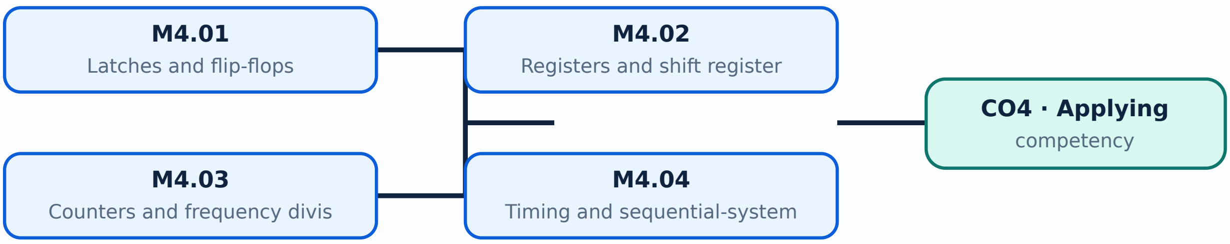
Malayalam support: M4.04 6 Applying counters using flip flops Series Test - II 1 □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

Learning goals

- Explain the role of a clock and state.
- Read characteristic and excitation tables.
- Select a register or counter arrangement.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Latches and flip-flops (3 hours)

Concept and explanation

A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Compare level and edge triggering.
- Use preset and clear safely.
- Read a characteristic table.

Handbook treatment

M4.01 — Latches and flip-flops (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.01 — Latches and flip-flops (3 hours) using the official terminology retained above.
- Organise the important elements of M4.01 — Latches and flip-flops (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.01 — Latches and flip-flops (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.01 — Latches and flip-flops (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.01 — Latches and flip-flops (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.01 — Latches and flip-flops (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.01 — Latches and flip-flops (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.01 — Latches and flip-flops (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 — Latches and flip-flops (3 hours)?
- How would you distinguish M4.01 — Latches and flip-flops (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.01 — Latches and flip-flops (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.01 — Latches and flip-flops (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.01 — Latches and flip-flops (3 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

M4.02 — Registers and shift registers (3 hours)

Concept and explanation

A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□.

Study points

- Identify data direction.
- Explain serial and parallel transfer.
- Relate a register to temporary storage.

> **Engineering / workplace application:** Serial communication and delay lines use shift registers.

Handbook treatment

M4.02 — Registers and shift registers (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.02 — Registers and shift registers (3 hours) using the official terminology retained above.
- Organise the important elements of M4.02 — Registers and shift registers (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.02 — Registers and shift registers (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.02 — Registers and shift registers (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.02 — Registers and shift registers (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.02 — Registers and shift registers (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.02 — Registers and shift registers (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.02 — Registers and shift registers (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 — Registers and shift registers (3 hours)?
- How would you distinguish M4.02 — Registers and shift registers (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.02 — Registers and shift registers (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 — Registers and shift registers (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.02 — Registers and shift registers (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M4.03 — Counters and frequency division (3 hours)

Concept and explanation

Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Distinguish asynchronous and synchronous counters.
- Determine modulus from the state sequence.
- Explain reset and unused states.

Handbook treatment

M4.03 — Counters and frequency division (3 hours) must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

Learning objectives

- Define and scope M4.03 — Counters and frequency division (3 hours) using the official terminology retained above.
- Organise the important elements of M4.03 — Counters and frequency division (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.03 — Counters and frequency division (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.03 — Counters and frequency division (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.03 — Counters and frequency division (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.
- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

Engineering application lens

In an engineering setting, M4.03 — Counters and frequency division (3 hours) is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

Worked answer method

Given: the quantities and conditions stated in the question.
Required: the unknown quantity, including its unit.
Calculation: write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
Answer: report the result with a suitable number of significant figures and one sentence of interpretation.

Exam answer blueprint

For a short-answer question on M4.03 — Counters and frequency division (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.03 — Counters and frequency division (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 — Counters and frequency division (3 hours)?
- How would you distinguish M4.03 — Counters and frequency division (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.03 — Counters and frequency division (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 — Counters and frequency division (3 hours) incorrect?

Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

Malayalam support: M4.03 — Counters and frequency division (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

M4.04 — Timing and sequential-system design (3 hours)

Concept and explanation

A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

> **Malayalam support:** [illegible] [illegible] [illegible] English technical terms [illegible] [illegible].

Study points

- Draw a state diagram.
- Check timing constraints.
- Use a reset state for predictable startup.

Handbook treatment

M4.04 — Timing and sequential-system design (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.04 — Timing and sequential-system design (3 hours) using the official terminology retained above.
- Organise the important elements of M4.04 — Timing and sequential-system design (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.04 — Timing and sequential-system design (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Sequential circuits.
- Write an examination answer on M4.04 — Timing and sequential-system design (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.04 — Timing and sequential-system design (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.04 — Timing and sequential-system design (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.
Required: definition, explanation, comparison, procedure, calculation, or application as requested.
Method: organise the answer under principle, named elements, sequence or relationship, and application.
Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.04 — Timing and sequential-system design (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.04 — Timing and sequential-system design (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 — Timing and sequential-system design (3 hours)?
- How would you distinguish M4.04 — Timing and sequential-system design (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.04 — Timing and sequential-system design (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 — Timing and sequential-system design (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.04 — Timing and sequential-system design (3 hours) □□□□ topic □□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

Module summary: Sequential design requires attention to timing, state transitions, setup, hold and reset behaviour.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

Open the module to view its labelled learning-map diagram.

Module 2: Boolean algebra

Open the module to view its labelled learning-map diagram.

Module 3: Combinational

Open the module to view its labelled learning-map diagram.

Module 4: Sequential

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- No separate activity list is provided in the supplied PDF.

Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Series Test: 2 periods/assessment component as stated in the supplied syllabus; the supplied header does not state a separate CIE/ESE mark distribution.

Module-wise Questions and Answers

module-1 — Number systems and digital codes

1. Explain Number systems and base conversion. **[3 marks]**

A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Complements and binary arithmetic. **[3 marks]**

One’s complement and two’s complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Binary codes and data representation. **[3 marks]**

BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Boolean algebra and Karnaugh maps

4. Explain Boolean laws and logic gates. **[3 marks]**

Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan’s theorems are especially useful for changing a circuit between active-high and active-low forms.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

5. Explain Canonical forms, minterms and maxterms. **[3 marks]**

A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Karnaugh-map simplification. **[3 marks]**

A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Combinational circuits

7. Explain Half and full adders. **[3 marks]**

A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Subtractors and arithmetic circuits. **[3 marks]**

Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

9. Explain Encoders, decoders and comparators. **[3 marks]**

An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Multiplexers and demultiplexers. **[3 marks]**

A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Sequential circuits

11. Explain Latches and flip-flops. **[3 marks]**

A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Registers and shift registers. **[3 marks]**

A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

13. Explain Counters and frequency division. **[3 marks]**

Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Timing and sequential-system design. **[3 marks]**

A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Number systems and base conversion* with one relevant application. (5 marks)
2. **2.** Define or explain *Complements and binary arithmetic* with one relevant application. (5 marks)
3. **3.** Define or explain *Boolean laws and logic gates* with one relevant application. (5 marks)
4. **4.** Define or explain *Canonical forms, minterms and maxterms* with one relevant application. (5 marks)
5. **5.** Define or explain *Half and full adders* with one relevant application. (5 marks)
6. **6.** Define or explain *Subtractors and arithmetic circuits* with one relevant application. (5 marks)
7. **7.** Define or explain *Latches and flip-flops* with one relevant application. (5 marks)
8. **8.** Define or explain *Registers and shift registers* with one relevant application. (5 marks)

> **Writing advice:** Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Number systems and digital codes:** A reliable conversion method prevents errors when a digital quantity is represented in another base.
- **Boolean algebra and Karnaugh maps:** Simplification reduces gate count, delay and wiring in a practical digital circuit.
- **Combinational circuits:** A clean design proceeds from specification to truth table, simplification, circuit drawing and verification.
- **Sequential circuits:** Sequential design requires attention to timing, state transitions, setup, hold and reset behaviour.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.

- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Number systems and base conversion	A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.
Complements and binary arithmetic	One's complement and two's complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.
Binary codes and data representation	BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.
Boolean laws and logic gates	Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan's theorems are especially useful for changing a circuit between active-high and active-low forms.
Canonical forms, minterms and maxterms	A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.
Karnaugh-map simplification	A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.
Half and full adders	A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.
Subtractors and arithmetic circuits	Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.
Encoders, decoders and comparators	An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.
Multiplexers and demultiplexers	A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.

Term	Short meaning
Latches and flip-flops	A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.
Registers and shift registers	A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.
Counters and frequency division	Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.
Timing and sequential-system design	A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Official References and Resources

References listed in the supplied syllabus

1. Digital Design, M. Morris Mano.
2. Fundamentals of Digital Circuits, A. Anand Kumar.

Official learning resources listed

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In-page Search